Three-axis estimator in the measurement coordinate

2026-10-23 states in (x, y, z): tracking error and the motion gain matrix $\bar{\mathbf{A}}$; wall angles computed from $\hat{\bar{\mathbf{A}}}$

Part A: one step of the algorithm (order of the notes). Part B: the key derivation steps. Q, R, $\mathbf{P}[0]$ and initial values are not written here.

Part A One step of the algorithm

A1 Input and measurement

$$\bar{\mathbf{p}}_d[k], \quad \bar{\mathbf{p}}_d[k+1]$$

$$\bar{\mathbf{p}}_m[k]$$

$$\delta\bar{\mathbf{p}}_m[k] = \bar{\mathbf{p}}_d[k-d] - \bar{\mathbf{p}}_m[k]$$

A2 Control law

$$\hat{\bar{\mathbf{p}}}[k] = \bar{\mathbf{p}}_m[k] + \sum_{i=1}^{d} \hat{\bar{\mathbf{A}}}[k-i]\,\bar{\mathbf{f}}_d[k-i]$$

$$\bar{\mathbf{f}}_d[k] = \hat{\bar{\mathbf{A}}}^{-1}[k]\left\{\bar{\mathbf{p}}_d[k+1] - \hat{\bar{\mathbf{p}}}[k] - \lambda_c\left(\bar{\mathbf{p}}_d[k] - \hat{\bar{\mathbf{p}}}[k]\right)\right\}$$

A3 Deterministic and random components, measured motion gain matrix

$$\overline{\delta\bar{\mathbf{p}}_m}[k+1] = a_{\mathrm{pd}}\,\delta\bar{\mathbf{p}}_m[k] + (1-a_{\mathrm{pd}})\,\overline{\delta\bar{\mathbf{p}}_m}[k]$$

$$\delta\bar{\mathbf{p}}_r[k] = \delta\bar{\mathbf{p}}_m[k] - \overline{\delta\bar{\mathbf{p}}_m}[k]$$

$$\bar{\mathbf{A}}_m[k] = \frac{\delta\bar{\mathbf{p}}_r[k]\,\delta\bar{\mathbf{p}}_r^{T}[k] - C_n\,\boldsymbol{\Sigma}_n}{C_T\,\kappa_T}$$

$$\bar{\mathbf{a}}_m[k] = \left[\,[\bar{\mathbf{A}}_m]_{xx} \;\; [\bar{\mathbf{A}}_m]_{yy} \;\; [\bar{\mathbf{A}}_m]_{zz} \;\; [\bar{\mathbf{A}}_m]_{xy} \;\; [\bar{\mathbf{A}}_m]_{xz} \;\; [\bar{\mathbf{A}}_m]_{yz}\,\right]^{T}$$

A4 Estimation error and state update

$$\mathbf{y}[k] = \begin{bmatrix}\delta\bar{\mathbf{p}}_m[k]\\ \bar{\mathbf{a}}_m[k]\end{bmatrix}, \qquad \hat{\mathbf{y}}[k] = \begin{bmatrix}\delta\hat{\bar{\mathbf{p}}}_1[k]\\ \hat{\bar{\mathbf{a}}}[k]\end{bmatrix}, \qquad \mathbf{e}_y[k] = \mathbf{y}[k] - \hat{\mathbf{y}}[k]$$

$$\mathbf{L}[k] = \mathbf{P}^{f}[k]\,\mathbf{H}^{T}\left(\mathbf{H}\,\mathbf{P}^{f}[k]\,\mathbf{H}^{T} + \mathbf{R}[k]\right)^{-1}$$

$$\delta\hat{\bar{\mathbf{p}}}_1[k+1] = \delta\hat{\bar{\mathbf{p}}}_2[k] + \mathbf{L}_1[k]\,\mathbf{e}_y[k]$$

$$\delta\hat{\bar{\mathbf{p}}}_2[k+1] = \delta\hat{\bar{\mathbf{p}}}_3[k] + \mathbf{L}_2[k]\,\mathbf{e}_y[k]$$

$$\delta\hat{\bar{\mathbf{p}}}_3[k+1] = \lambda_c\,\delta\hat{\bar{\mathbf{p}}}_3[k] + \mathbf{L}_3[k]\,\mathbf{e}_y[k]$$

$$\hat{\bar{\mathbf{a}}}[k+1] = \hat{\bar{\mathbf{a}}}[k] + \mathbf{L}_4[k]\,\mathbf{e}_y[k]$$

$\mathbf{L}_1, \mathbf{L}_2, \mathbf{L}_3$: 3×9; $\mathbf{L}_4$: 6×9. $\hat{\bar{\mathbf{A}}}[k+1]$ is the symmetric matrix built from $\hat{\bar{\mathbf{a}}}[k+1]$.

A5 Wall angles from the estimated motion gain matrix

$$
\begin{aligned}
t[k+1] &= \hat{\bar{a}}_{xx} + \hat{\bar{a}}_{yy} + \hat{\bar{a}}_{zz} \\
s[k+1] &= \hat{\bar{a}}_{xx}^2 + \hat{\bar{a}}_{yy}^2 + \hat{\bar{a}}_{zz}^2 + 2\,\hat{\bar{a}}_{xy}^2 + 2\,\hat{\bar{a}}_{xz}^2 + 2\,\hat{\bar{a}}_{yz}^2 \\
\hat{\bar{a}}_T[k+1] &= \frac{2\,t[k+1] + \sqrt{6\,s[k+1] - 2\,t^2[k+1]}}{6} \\
\hat{\bar{a}}_N[k+1] &= t[k+1] - 2\,\hat{\bar{a}}_T[k+1] \\
\hat{\mathbf{w}}[k+1] &= \pm\,\frac{\mathbf{c}}{\|\mathbf{c}\|}, \qquad \mathbf{c} = \begin{bmatrix} \hat{\bar{a}}_{xz} \\ \hat{\bar{a}}_{yz} \\ \hat{\bar{a}}_{zz} - \hat{\bar{a}}_T[k+1] \end{bmatrix}, \qquad \hat{w}_z > 0 \\
\hat{\theta}_x[k+1] &= -\arcsin \hat{w}_y \\
\hat{\theta}_y[k+1] &= \operatorname{atan2}\big(\hat{w}_x,\ \hat{w}_z\big) \\
\hat{\mathbf{V}}[k+1] &= \mathbf{R}_y\big(\hat{\theta}_y[k+1]\big)\,\mathbf{R}_x\big(\hat{\theta}_x[k+1]\big) = [\,\hat{\mathbf{u}}\ \ \hat{\mathbf{v}}\ \ \hat{\mathbf{w}}\,]
\end{aligned}
$$

$\hat{\mathbf{V}}$ is used only here and for commands along $\hat{\mathbf{w}}$, $\hat{\mathbf{u}}$; the estimator (A1–A4, A6) does not use it.

A6 Error covariance

$$
\begin{aligned}
\mathbf{P}[k] &= \big(\mathbf{I} - \mathbf{L}[k]\,\mathbf{H}\big)\,\mathbf{P}^f[k] \\
\mathbf{P}^f[k+1] &= \mathbf{F}_e[k]\,\mathbf{P}[k]\,\mathbf{F}_e^T[k] + \mathbf{Q}[k]
\end{aligned}
$$

Part B Key derivation steps

1 Motion gain matrix of a tilted wall

$$\mathbf{V} = \mathbf{R}_y(\theta_y)\,\mathbf{R}_x(\theta_x) = [\,\mathbf{u}\;\;\mathbf{v}\;\;\mathbf{w}\,]$$

$$\mathbf{w} = \begin{bmatrix} \sin\theta_y \cos\theta_x \\ -\sin\theta_x \\ \cos\theta_y \cos\theta_x \end{bmatrix}$$

$$\bar{\mathbf{\Lambda}} = \mathrm{diag}(\bar{a}_T,\ \bar{a}_T,\ \bar{a}_N)$$

$$\bar{\mathbf{A}} = \mathbf{V}\,\bar{\mathbf{\Lambda}}\,\mathbf{V}^T = \bar{a}_T\,\mathbf{I} + (\bar{a}_N - \bar{a}_T)\,\mathbf{w}\,\mathbf{w}^T$$

$$\bar{\mathbf{\Lambda}} = \mathbf{V}^T\,\bar{\mathbf{A}}\,\mathbf{V}$$

2 The wall angles are the off-diagonal entries of $\bar{\mathbf{A}}$

Small angles: drop every product of two angles.

$$\mathbf{w} \approx \begin{bmatrix} \theta_y \\ -\theta_x \\ 1 \end{bmatrix}$$

$$\mathbf{w}\,\mathbf{w}^T \approx \begin{bmatrix} 0 & 0 & \theta_y \\ 0 & 0 & -\theta_x \\ \theta_y & -\theta_x & 1 \end{bmatrix}$$

$$\bar{\mathbf{A}} \approx \begin{bmatrix} \bar{a}_T & 0 & -(\bar{a}_T - \bar{a}_N)\,\theta_y \\ 0 & \bar{a}_T & (\bar{a}_T - \bar{a}_N)\,\theta_x \\ -(\bar{a}_T - \bar{a}_N)\,\theta_y & (\bar{a}_T - \bar{a}_N)\,\theta_x & \bar{a}_N \end{bmatrix}$$

$$\left[\bar{\mathbf{A}}\right]_{xz} \approx -(\bar{a}_T - \bar{a}_N)\,\theta_y$$

$$\left[\bar{\mathbf{A}}\right]_{yz} \approx (\bar{a}_T - \bar{a}_N)\,\theta_x$$

$$\left[\bar{\mathbf{A}}\right]_{xy} \approx 0$$

3 Process: control law and tracking error

$$
\begin{aligned}
\bar{\mathbf{p}}[k+1] &= \bar{\mathbf{p}}[k] + \bar{\mathbf{A}}\left(\bar{\mathbf{f}}_d[k] + \bar{\mathbf{f}}_T[k]\right) \\
\delta\bar{\mathbf{p}}[k] &= \bar{\mathbf{p}}_d[k] - \bar{\mathbf{p}}[k] \\
\bar{\mathbf{p}}_m[k] &= \bar{\mathbf{p}}[k-d] + \mathbf{n}_p[k] \\
\delta\bar{\mathbf{p}}_m[k] &= \bar{\mathbf{p}}_d[k-d] - \bar{\mathbf{p}}_m[k] \\
\hat{\bar{\mathbf{p}}}[k] &= \bar{\mathbf{p}}_m[k] + \sum_{i=1}^{d} \hat{\bar{\mathbf{A}}}[k-i]\,\bar{\mathbf{f}}_d[k-i] \\
\bar{\mathbf{f}}_d[k] &= \hat{\bar{\mathbf{A}}}^{-1}[k]\left\{\bar{\mathbf{p}}_d[k+1] - \hat{\bar{\mathbf{p}}}[k] - \lambda_c\left(\bar{\mathbf{p}}_d[k] - \hat{\bar{\mathbf{p}}}[k]\right)\right\} \\
\delta\bar{\mathbf{p}}[k+1] &= \lambda_c\,\delta\bar{\mathbf{p}}[k] - \left(\bar{\mathbf{A}} - \hat{\bar{\mathbf{A}}}[k]\right)\bar{\mathbf{F}}_{dp}[k] - \boldsymbol{\varepsilon}_p[k] \\
\bar{\mathbf{F}}_{dp}[k] &= \bar{\mathbf{f}}_d[k] + (1-\lambda_c)\sum_{i=1}^{d} \bar{\mathbf{f}}_d[k-i]
\end{aligned}
$$

True state of the process

$$
\begin{aligned}
\delta\bar{\mathbf{p}}_1[k+1] &= \delta\bar{\mathbf{p}}_2[k] \\
\delta\bar{\mathbf{p}}_2[k+1] &= \delta\bar{\mathbf{p}}_3[k] \\
\delta\bar{\mathbf{p}}_3[k+1] &= \lambda_c\,\delta\bar{\mathbf{p}}_3[k] - \left(\bar{\mathbf{A}} - \hat{\bar{\mathbf{A}}}[k]\right)\bar{\mathbf{F}}_{dp}[k] - \boldsymbol{\varepsilon}_p[k] \\
\bar{\mathbf{a}}[k+1] &= \bar{\mathbf{a}}[k] + \mathbf{w}_a[k]
\end{aligned}
$$

4 State vector and estimator

$$
\begin{aligned}
\bar{\mathbf{a}} &= \begin{bmatrix} \bar{a}_{xx} & \bar{a}_{yy} & \bar{a}_{zz} & \bar{a}_{xy} & \bar{a}_{xz} & \bar{a}_{yz} \end{bmatrix}^T \\
\mathbf{x}[k] &= \begin{bmatrix} \delta\bar{\mathbf{p}}_1^T[k] & \delta\bar{\mathbf{p}}_2^T[k] & \delta\bar{\mathbf{p}}_3^T[k] & \bar{\mathbf{a}}^T[k] \end{bmatrix}^T \qquad (15\times 1,\ d=2)
\end{aligned}
$$

$$
\begin{aligned}
\delta\hat{\bar{\mathbf{p}}}_1[k+1] &= \delta\hat{\bar{\mathbf{p}}}_2[k] + \mathbf{L}_1[k]\,\mathbf{e}_y[k] \\
\delta\hat{\bar{\mathbf{p}}}_2[k+1] &= \delta\hat{\bar{\mathbf{p}}}_3[k] + \mathbf{L}_2[k]\,\mathbf{e}_y[k] \\
\delta\hat{\bar{\mathbf{p}}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{\mathbf{p}}}_3[k] + \mathbf{L}_3[k]\,\mathbf{e}_y[k] \\
\hat{\bar{\mathbf{a}}}[k+1] &= \hat{\bar{\mathbf{a}}}[k] + \mathbf{L}_4[k]\,\mathbf{e}_y[k]
\end{aligned}
$$

5 Dynamics of estimation errors

$\mathbf{e}_{(\cdot)} = (\cdot) - (\hat{\cdot})$; $\mathbf{e}_a = \bar{\mathbf{a}} - \hat{\bar{\mathbf{a}}} = [\, e_{xx}\ e_{yy}\ e_{zz}\ e_{xy}\ e_{xz}\ e_{yz}\,]^T$; F_x, F_y, F_z = components of $\bar{\mathbf{F}}_{dp}[k]$

$$\bar{\mathbf{A}} - \hat{\bar{\mathbf{A}}} = \begin{bmatrix} e_{xx} & e_{xy} & e_{xz} \\ e_{xy} & e_{yy} & e_{yz} \\ e_{xz} & e_{yz} & e_{zz} \end{bmatrix}$$

$$\left(\bar{\mathbf{A}} - \hat{\bar{\mathbf{A}}}\right) \bar{\mathbf{F}}_{dp} = \begin{bmatrix} e_{xx}\,F_x + e_{xy}\,F_y + e_{xz}\,F_z \\ e_{xy}\,F_x + e_{yy}\,F_y + e_{yz}\,F_z \\ e_{xz}\,F_x + e_{yz}\,F_y + e_{zz}\,F_z \end{bmatrix} = \begin{bmatrix} F_x & 0 & 0 & F_y & F_z & 0 \\ 0 & F_y & 0 & F_x & 0 & F_z \\ 0 & 0 & F_z & 0 & F_x & F_y \end{bmatrix} \mathbf{e}_a$$

$$\mathbf{e}_1[k+1] = \mathbf{e}_2[k]$$

$$\mathbf{e}_2[k+1] = \mathbf{e}_3[k]$$

$$\mathbf{e}_3[k+1] = \lambda_c\,\mathbf{e}_3[k] - \begin{bmatrix} F_x & 0 & 0 & F_y & F_z & 0 \\ 0 & F_y & 0 & F_x & 0 & F_z \\ 0 & 0 & F_z & 0 & F_x & F_y \end{bmatrix} \mathbf{e}_a[k] - \boldsymbol{\varepsilon}_p[k]$$

$$\mathbf{e}_a[k+1] = \mathbf{e}_a[k] + \mathbf{w}_a[k]$$

$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k] - \mathbf{L}[k]\,\mathbf{H}\,\mathbf{e}[k] + \mathbf{q}[k]$$

$$\mathbf{F}_e[k] = \begin{bmatrix} \mathbf{0} & \mathbf{I}_3 & \mathbf{0} & & & \mathbf{0} & & \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_3 & & & \mathbf{0} & & \\ \mathbf{0} & \mathbf{0} & \lambda_c\,\mathbf{I}_3 & - & \begin{bmatrix} F_x & 0 & 0 & F_y & F_z & 0 \\ 0 & F_y & 0 & F_x & 0 & F_z \\ 0 & 0 & F_z & 0 & F_x & F_y \end{bmatrix} & & \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & & & \mathbf{I}_6 & & \end{bmatrix}$$

$$\mathbf{q}[k] = \begin{bmatrix} \mathbf{0}^T & \mathbf{0}^T & -\boldsymbol{\varepsilon}_p^T[k] & \mathbf{w}_a^T[k] \end{bmatrix}^T$$

6 Process outputs and output matrix

$$\mathrm{E}\big[\delta\bar{\mathbf{p}}_r\,\delta\bar{\mathbf{p}}_r^T\big] = C_T\,\kappa_T\,\bar{\mathbf{A}} + C_n\,\boldsymbol{\Sigma}_n$$
$$\mathrm{E}\big[\bar{\mathbf{A}}_m\big] = \bar{\mathbf{A}}$$
$$\mathrm{E}\big[\bar{\mathbf{a}}_m\big] = \bar{\mathbf{a}}$$

$$\mathbf{y}[k] = \begin{bmatrix}\delta\bar{\mathbf{p}}_m[k]\\ \bar{\mathbf{a}}_m[k]\end{bmatrix}, \qquad \hat{\mathbf{y}}[k] = \begin{bmatrix}\delta\hat{\bar{\mathbf{p}}}_1[k]\\ \hat{\bar{\mathbf{a}}}[k]\end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix}\mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I}_6\end{bmatrix}$$

Two paths into $\mathbf{e}_a$

$$\frac{\partial\,\delta\bar{\mathbf{p}}_3[k+1]}{\partial\,\bar{\mathbf{a}}} = -\begin{bmatrix}F_x & 0 & 0 & F_y & F_z & 0\\ 0 & F_y & 0 & F_x & 0 & F_z\\ 0 & 0 & F_z & 0 & F_x & F_y\end{bmatrix} \qquad \text{(motion)}$$

$$\frac{\partial\,\bar{\mathbf{a}}_m}{\partial\,\bar{\mathbf{a}}} = \mathbf{I}_6 \qquad \text{(random component)}$$

7 Wall angles from $\hat{\bar{\mathbf{A}}}$

Two invariants of $\bar{\mathbf{A}}$ give the two gains; the third column of $\bar{\mathbf{A}} - \bar{a}_T\,\mathbf{I}$ points along $\mathbf{w}$.

$$\bar{\mathbf{A}} = \bar{a}_T\,\mathbf{I} + (\bar{a}_N - \bar{a}_T)\,\mathbf{w}\,\mathbf{w}^T$$

$$t = \operatorname{tr}\bar{\mathbf{A}} = 2\,\bar{a}_T + \bar{a}_N$$

$$s = \sum_{i,j}\big[\bar{\mathbf{A}}\big]_{ij}^2 = 2\,\bar{a}_T^2 + \bar{a}_N^2$$

$$6\,s - 2\,t^2 = 4\,(\bar{a}_T - \bar{a}_N)^2$$

$$\bar{a}_T = \frac{2\,t + \sqrt{6\,s - 2\,t^2}}{6} \qquad (\bar{a}_T > \bar{a}_N)$$

$$\bar{a}_N = t - 2\,\bar{a}_T$$

$$\big(\bar{\mathbf{A}} - \bar{a}_T\,\mathbf{I}\big)\,\mathbf{e}_z = (\bar{a}_N - \bar{a}_T)\,w_z\,\mathbf{w}$$

$$\big(\bar{\mathbf{A}} - \bar{a}_T\,\mathbf{I}\big)\,\mathbf{e}_z = \begin{bmatrix} \bar{a}_{xz} \\ \bar{a}_{yz} \\ \bar{a}_{zz} - \bar{a}_T \end{bmatrix}$$

$$\mathbf{w} = \pm\,\frac{\big(\bar{\mathbf{A}} - \bar{a}_T\,\mathbf{I}\big)\,\mathbf{e}_z}{\big\|\big(\bar{\mathbf{A}} - \bar{a}_T\,\mathbf{I}\big)\,\mathbf{e}_z\big\|}$$

$$\theta_x = -\arcsin w_y$$

$$\theta_y = \operatorname{atan2}\big(w_x,\ w_z\big)$$

Appendix: incremental small-angle form (not used)

$\hat{\bar{\mathbf{A}}}[k+1]$ written as a wall at the angles $\hat{\theta}_x[k]+\delta_x$, $\hat{\theta}_y[k]+\delta_y$; first order in δ_x, δ_y.

$$\hat{\bar{\mathbf{A}}}[k+1] = \mathbf{V}\big(\hat{\boldsymbol{\theta}}[k]+\boldsymbol{\delta}\big)\,\mathrm{diag}\big(\hat{\bar{a}}_T,\ \hat{\bar{a}}_T,\ \hat{\bar{a}}_N\big)\,\mathbf{V}^T\big(\hat{\boldsymbol{\theta}}[k]+\boldsymbol{\delta}\big)$$

$$\frac{\partial \mathbf{V}}{\partial \theta_x} = \mathbf{V}\,[\mathbf{e}_x]_\times$$

$$\frac{\partial \mathbf{V}}{\partial \theta_y} = [\mathbf{e}_y]_\times\,\mathbf{V}$$

$$\hat{\mathbf{V}}^T[k]\,\mathbf{e}_y = \begin{bmatrix} 0 \\ \cos\hat{\theta}_x[k] \\ -\sin\hat{\theta}_x[k] \end{bmatrix}$$

$$\hat{\mathbf{V}}^T[k]\,\mathbf{V}\big(\hat{\boldsymbol{\theta}}[k]+\boldsymbol{\delta}\big) \approx \mathbf{I} + [\boldsymbol{\omega}]_\times$$

$$\boldsymbol{\omega} = \begin{bmatrix} \delta_x \\ \cos\hat{\theta}_x[k]\,\delta_y \\ -\sin\hat{\theta}_x[k]\,\delta_y \end{bmatrix}$$

$${}^{v}\hat{\bar{\mathbf{A}}}[k+1] = \hat{\mathbf{V}}^T[k]\,\hat{\bar{\mathbf{A}}}[k+1]\,\hat{\mathbf{V}}[k]$$

$$\approx \mathrm{diag}\big(\hat{\bar{a}}_T,\ \hat{\bar{a}}_T,\ \hat{\bar{a}}_N\big) + [\boldsymbol{\omega}]_\times\,\mathrm{diag}\big(\hat{\bar{a}}_T,\ \hat{\bar{a}}_T,\ \hat{\bar{a}}_N\big) - \mathrm{diag}\big(\hat{\bar{a}}_T,\ \hat{\bar{a}}_T,\ \hat{\bar{a}}_N\big)\,[\boldsymbol{\omega}]_\times$$

$$\big[{}^{v}\hat{\bar{\mathbf{A}}}[k+1]\big]_{vw} \approx (\hat{\bar{a}}_T - \hat{\bar{a}}_N)\,\delta_x$$

$$\big[{}^{v}\hat{\bar{\mathbf{A}}}[k+1]\big]_{uw} \approx -(\hat{\bar{a}}_T - \hat{\bar{a}}_N)\,\cos\hat{\theta}_x[k]\,\delta_y$$

$$\big[{}^{v}\hat{\bar{\mathbf{A}}}[k+1]\big]_{uv} \approx 0$$

$$\delta_x \approx \frac{\big[{}^{v}\hat{\bar{\mathbf{A}}}[k+1]\big]_{vw}}{\hat{\bar{a}}_T - \hat{\bar{a}}_N}$$

$$\delta_y \approx -\frac{\big[{}^{v}\hat{\bar{\mathbf{A}}}[k+1]\big]_{uw}}{(\hat{\bar{a}}_T - \hat{\bar{a}}_N)\,\cos\hat{\theta}_x[k]}$$

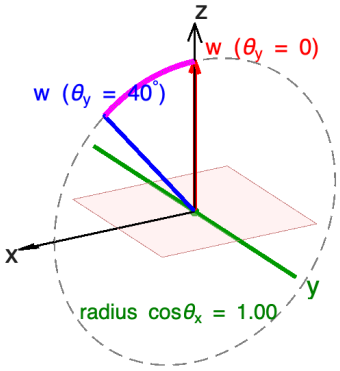
$$\hat{\boldsymbol{\theta}}[k+1] = \hat{\boldsymbol{\theta}}[k] + \boldsymbol{\delta} = \boldsymbol{\theta}\big(\hat{\bar{\mathbf{A}}}[k+1]\big) + O\big(|\boldsymbol{\delta}|^2\big)$$

Every eigenvector of $\hat{\bar{\mathbf{A}}}$ is a fixed point

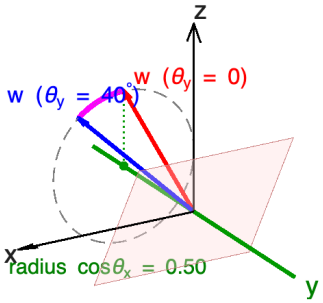
$$\hat{\mathbf{V}}[k] = [\,\mathbf{q}_1\ \ \mathbf{q}_2\ \ \mathbf{q}_3\,],\quad \hat{\bar{\mathbf{A}}}\,\mathbf{q}_i = \mu_i\,\mathbf{q}_i \ \Rightarrow\ \big[{}^{v}\hat{\bar{\mathbf{A}}}\big]_{vw} = \big[{}^{v}\hat{\bar{\mathbf{A}}}\big]_{uw} = 0 \ \Rightarrow\ \boldsymbol{\delta} = \mathbf{0}$$

$$\hat{\mathbf{w}}[k] \parallel \text{eigenvector of } \bar{a}_T \ \Rightarrow\ \frac{\big[{}^{v}\hat{\bar{\mathbf{A}}}\big]_{uu} + \big[{}^{v}\hat{\bar{\mathbf{A}}}\big]_{vv}}{2} - \big[{}^{v}\hat{\bar{\mathbf{A}}}\big]_{ww} = \frac{\bar{a}_T + \bar{a}_N}{2} - \bar{a}_T = -\frac{\bar{a}_T - \bar{a}_N}{2}$$

$\theta_x = 0^\circ$: $\Delta\theta_y = 40^\circ$ → w turns 40.0°



$\theta_x = 60^\circ$: $\Delta\theta_y = 40^\circ$ → w turns 19.7°



Comparison

	Notes (Word)	1015 (angles as states)	This document
States	${}^{V}\delta\mathbf{p}$, $\boldsymbol{\lambda}$, $\boldsymbol{\theta}$ (local)	$\delta\bar{\mathbf{p}}$ (x, y, z), $\bar{a}_T$, $\bar{a}_N$, θ_x, θ_y	$\delta\bar{\mathbf{p}}$, $\bar{\mathbf{a}}$ (x, y, z)
Gain matrix	$\boldsymbol{\lambda}$ with $\mathbf{V}$	$\hat{\mathbf{V}}\,\mathrm{diag}(\hat{\bar{a}}_T, \hat{\bar{a}}_T, \hat{\bar{a}}_N)\,\hat{\mathbf{V}}^T$	six entries of $\hat{\bar{\mathbf{A}}}$
Angles	states, measured θ_m	states, rows of ${}^{v}\bar{\mathbf{A}}_m$	computed from $\hat{\bar{\mathbf{A}}}$ (A5, closed form)
End of step	$\mathbf{V}[k]$ out, $\mathbf{V}^T[k+1]$ in	none	none
$\mathbf{F}_e$	$\mathbf{G}_\lambda$, $\mathbf{G}_\theta$	gain and angle columns, $\cos\hat{\theta}_x$	force matrix only
$\mathbf{H}$	selects ${}^{V}\delta\mathbf{p}_1$, $\boldsymbol{\lambda}$, $\boldsymbol{\theta}$	angle rows $\hat{\bar{a}}_T - \hat{\bar{a}}_N$	$\mathbf{I}_3$, $\mathbf{I}_6$